

Skills Summary

Convergence Tests and the Endpoints of an Interval

Use this reference while completing your worksheet or when studying.

Skill 1 — The n th-term test goes one way

If $\lim a_n \neq 0$, then $\sum a_n$ diverges. That is the whole test.

- A limit of 0 licenses **nothing**. It is necessary for convergence, never sufficient — $\sum 1/n$ is the standing counterexample.
- So compute the limit first: a non-zero answer ends the problem, and a zero answer means the real work has not started.
- For a ratio of polynomials, divide through by the highest power of n .

Example: Apply the n th-term test to the series with $a_n = \frac{5n-2}{3n+4}$.

Step 1. The test only needs the limit of the terms, so compute that before reaching for any comparison — if it is non-zero the question is already answered.

Step 2. Divide numerator and denominator by n : $\frac{5-2/n}{3+4/n}$, and let $n \rightarrow \infty$.

The limit is $\frac{5}{3}$, which is not 0, so the series diverges.

Try it: Find $\lim_{n \rightarrow \infty} \frac{2n+9}{6n-1}$, then say what the n th-term test concludes.
check: $\frac{2}{3}$, so it diverges

Skill 2 — When the ratio test says nothing

The ratio test concludes only when its limit misses 1.

- $L < 1$ converges, $L > 1$ diverges, $L = 1$ is **inconclusive** — both $\sum 1/n$ and $\sum 1/n^2$ give $L = 1$ and behave differently.
- At $L = 1$, switch tools: p -series, integral test, telescoping, or a direct comparison.
- Geometric with $|r| < 1$: the sum is $\frac{a}{1-r}$, where a is the *first term the sum actually starts with*. Dividing by r instead of by $1-r$ is the classic slip.

Example: Find the exact sum of $\sum_{n=0}^{\infty} 5 \left(\frac{1}{4}\right)^n$.

Step 1. The terms have a constant ratio, so this is geometric and the closed-form sum applies — no test is needed beyond checking that the ratio is under 1 in size.

Step 2. The index starts at 0, so the first term is 5 and the ratio is $\frac{1}{4}$: the sum is $\frac{5}{1-1/4}$.

The sum is $\frac{20}{3}$.

Try it: Find the exact sum of $\sum_{n=0}^{\infty} 3 \left(\frac{2}{5}\right)^n$.

check: 5

Skill 3 — Endpoints of the interval

The ratio test gives the radius. The endpoints are yours to test.

- Endpoints are $c-r$ and $c+r$: the radius is a *distance* from the centre, never a coordinate.
- At either endpoint the ratio-test limit is exactly 1, so that test is guaranteed to be silent there. Substitute and run a different test.
- The two ends are independent — one may be included and the other excluded, so an interval like $[c-r, c+r)$ is entirely normal.

Example: A power series is centred at $c = -4$ with radius $r = 2$. Give the two endpoints of its interval of convergence.

Step 1. The radius measures distance from the centre, so step one radius to each side of the centre rather than reading the radius off as a position.

Step 2. Compute $c - r = -4 - 2$ and $c + r = -4 + 2$.

The endpoints are **-6** and **-2**, and each must now be substituted and tested on its own.

Try it: A power series is centred at $c = 7$ with radius $r = 5$. Give the two endpoints.

check: 2 and 12